

Year Two Lecture Notes

BSc Computer Science - Durham University

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Part I

Artificial Intelligence

1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	9th Oct	15th Dec	21st Jan	Dr Yang Long	50%
AI Search	15th Jan	18th Feb	18th Mar	Prof Iain Stewart	25%
Bias in AI	5th Feb	30th Mar	6th May	Dr Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	-
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

This section is taught by Dr Yang Long. The recommended reading for this topic is:

- Mathematics for Machine Learning (Deisenroth, M.P., Faisal, A.A., and Ong, C.S., 2020).
- Mastering Machine Learning with scikit-learn (Hackeling, G., 2017).

2.1 Introduction to Machine Learning

2.1.1 Module Information

This module is organised around a direct connection between teaching and practical work. The lectures introduce the concepts, mathematics, assumptions, and experimental consequences needed for that week's practical. The practical then produces the evidence used in the corresponding coursework report. This can be seen in Table 3.

Stage	What you do	What you keep
Lecture	Understand the main question, mechanisms, assumptions, and key equations.	Notes in your own words and questions that still need an answer.
Practical	Implement, test, compare, visualise, and discuss the method.	A notebook that runs from top to bottom with outputs included.
Weekly Report	Select the strongest evidence and explain why the result occurred.	A concise report that links theory, method, result, and interpretation.
Portfolio	Revise the nine reports and connect ideas across the module.	A coherent record of technical progress and developing judgement.

Table 3: The learning and coursework model.

For more information about expectations for evidence, see (Long and Durham University, 2026, pp.3-4).

2.1.2 What is Machine Learning?

Definition 2.1.2.1: Arthur Samuel said in 1959 that machine learning is the

“field of study that gives computers the ability to learn without being explicitly programmed.”

The field arose from the question – can a computer learn on its own in order to perform a specific task, rather than relying on a programmer explicitly setting up rules for it? Similarly, Tom Mitchell defined machine learning in 1997 as

“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”

From this, we can define a well-defined learning task as $\langle T, P, E \rangle$.

There are three typical types of machine learning, these are:

- **Definition 2.1.2.2:** Supervised learning seeks to learn the mapping (rules) between inputs and outputs. Labelled data (correct answers) is provided in past input and output pairs during the learning process to train the model on how it should behave for previously unseen data.
- **Definition 2.1.2.3:** Unsupervised learning seeks to discover the hidden pattern from a set of inputs. Unlabelled data is provided as the past input during the learning process. The program will then attempt to group similar instances into groups.
- **Definition 2.1.2.4:** Reinforcement learning works by providing occasional positive and negative feedback to reinforce behaviours. Good behaviours are rewarded with treats and therefore become more common whereas bad behaviours are punished and therefore become less common. This is a balance between exploration (of uncharted territory) and exploitation (of current knowledge).

2.1.3 Differences between AI, ML, and DL

Definition 2.1.3.1: Artificial intelligence is defined as human intelligence being exhibited by machines.

Definition 2.1.3.2: In simple terms, machine learning can be defined as an approach to achieve artificial intelligence.

Definition 2.1.3.3: Deep learning is defined as a technique for achieving machine learning. As such, we can consider AI to be a subset of ML and ML to be a subset of DL.

2.2 Machine Learning Workflow and Coding Environment

2.3 Data Understanding

2.4 Data Preparation Workflow

3 Artificial Intelligence Search

This section is taught by Professor Iain Stewart. There is no recommended reading for this topic however there may be an additional video to watch after lectures to enable more content to be covered.

Watch the FAQs video.

3.1 Coding Real-World Problems for Search Tree Expansion

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.2 Uninformed Search: BFS, DFS, and Iterative Deepening

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.3 General Heuristic Search: Greedy Best-First Search and A* Search

3.4 A* Search and Deriving Iterative Deepening A* Search

3.5 Heuristic Functions and Their Development

3.6 Local Search: Hill-Climbing, Simulated Annealing, and Genetic Algorithms

3.7 Swarm Behaviour, How Ants Forage, and a Simple S-ACO algorithm

3.8 Applying Different Ant Systems to the TSP

3.9 General Versions of Particle Swarm Optimisation and Differential Evolution

3.10 Applying Particle Swarm Optimization to the TSP

4 Ethics and Bias in AI

This section is taught by Dr Shauna Concannon. The recommended reading is handled on a weekly basis for this topic.

- 4.1 Introduction to AI Ethics and Related Concepts
- 4.2 Societal Impacts of AI
- 4.3 Defining Bias and Fairness in ML
- 4.4 How to Build Better Systems
- 4.5 Value Sensitive Design
- 4.6 Evaluations and Audits
- 4.7 Bias Mitigation Strategies
- 4.8 Transparency and Accountability for Trustworthy Systems
- 4.9 Explainable AI
- 4.10 Future Trends and Ethical Frontiers

Part II

Data Science

5 Module Introduction

Data Science has the module code COMP2271 and is led by Professor Thomas Erlebach.

5.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 4. Additional formatives are also shown in *italics*.

Title	Opening	Due	Feedback	Professor	Weight
Data Cleaning & Analytics	Week 11	-	11th Feb	Dr Igor Razgon	25%
Image Processing	15th Jan	2nd Mar	6th May	Dr George Koulieris	25%
Probability and Graphics	May Exam	-	-	Razgon & Koulieris	50%

Table 4: Assessments for Data Science.

Note: The Data Cleaning & Analytics test in Week 11 is closed book, and will be held during practicals (mine is Friday 4pm).

5.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 5.

Day	Time	Class Type	Location	Lecturer
Monday	12:00 - 13:00	Lecture	W103	Dr Igor Razgon
Friday	10:00 - 11:00	Lecture	W103	Dr Igor Razgon
Friday	16:00 - 18:00	Computer Class	CC-0007	-

Table 5: Michaelmas Term Timetable for Data Science.

6 Probability and Statistics

This section is taught by Dr Igor Razgon. The recommended reading for this topic is:

- Probability and Statistics (Degroot, M.H. and Schervish, M.J., 2012).

6.1 Probability and Statistics: Basic Concepts

6.1.1 Counting Theory

Definition 6.1.1.1: The product rule states that, if a procedure can be broken down into a sequence of two tasks with n_1 ways to do the first task and n_2 ways to do the second task, then there are $n_1 n_2$ ways to complete the entire procedure.

Example 6.1.1.1: John wants to fly to Canada from Germany with a connecting flight through the UK. There are 3 flights from Germany to the UK, and 2 flights from the UK to Canada. How many possible flight combinations are there?

Solution: $3 \times 2 = 6$.

□

Definition 6.1.1.2: The sum rule states that, if you consider two independent sub-tasks (we only have to do one), if the first can be done in n_1 ways, and the second can be done in n_2 ways, then the total number of ways to complete the task is $n_1 + n_2$.

Example 6.1.1.2: Mary wants to choose a student project from three lists: lecturer 1 offers 5 projects, lecturer 2 offers 2 projects, and lecturer 3 offers 10 projects. How many possible options for her?

Solution: $5 + 2 + 10 = 17$.

□

Definition 6.1.1.3: The inclusion-exclusion principle states that, if two tasks can be done at the same time (unlike the sum rule in which they are independent), then there is a possibility of overcounting since some ways may be counted twice. To counter this, we add up the number of ways to do each task, then subtract the number of ways to do both tasks.

$$|A \cup B| = (|A| + |B|) - (|A \cap B|) \quad (1)$$

$$|A \cup B \cup C| = (|A| + |B| + |C|) - (|A \cap B| + |A \cap C| + |B \cap C|) + (|A \cap B \cap C|) \quad (2)$$

Definition 6.1.1.4: An r-permutation of a set of r distinct objects is one ordered arrangement. We denote the number of r-permutations of a set with n elements in as $P(n, r)$.

$$P(n, r) = n \times (n - 1) \times (n - 2) \times \dots \times (n - r + 1) = \frac{n!}{(n - r)!} \quad (3)$$

Suppose that a set consists of n objects, of which n_1 are of one type (indistinguishable from one another), n_2 are of a second type, ..., and n_k are of a k_{th} type where $n = n_1 + n_2 + \dots + n_k$. The number of permutations of the objects can be calculated with (4).

$$P(n; n_1, n_2, \dots, n_k) = \frac{n!}{n_1! n_2! \dots n_k!} \quad (4)$$

Definition 6.1.1.5: A r-combination is a counting of distinct objects in an unordered arrangement of length r . The number of r combinations of a set with n distinct elements is given by (5).

$$C(n, r) = \binom{n}{r} = \frac{n!}{(n - r)! \times r!} \quad (5)$$

Example 6.1.1.3: Assume elements A_1, A_2, A_3, A_4 are in a set. We want to choose 2 elements from them.

Solution:

- If the order of elements is important: $P(4, 2) = \frac{4!}{(4-2)!} = 12$.
- If the order of elements is not important: $C(4, 2) = \frac{4!}{(4-2)!2!} = 6$.

□

6.1.2 Probability Theory

Definition 6.1.2.1: Probability is the formalisation of uncertain events. It enables one to infer probabilities of interest based on assumptions and observations.

Definition 6.1.2.2: Statistics is the practice of the science of collecting and analysing data. It deals with experimental data and extracting knowledge from it.

Definition 6.1.2.3: An experiment is a procedure that yields one of a given set of possible outcomes.

Definition 6.1.2.4: The sample space of the experiment is the set of possible outcomes.

Definition 6.1.2.5: An event is any subset of the sample space.

Theorem 6.1.2.1: The Theory of Probability states that, given a finite sample space of all outcomes S , assuming that all outcomes are equally likely, for an event set E where $E \subseteq S$, then the probability of occurrence of the event is given by (6).

$$P(E) = \frac{|E|}{|S|} \quad (6)$$

Proof: Proof omitted.

□

Definition 6.1.2.6: The axioms of probability are as follows:

- $P(E)$ is always between 0 and 1: $0 \leq P(E) \leq 1$.
- $P(S) = 1$.
- For disjoint events $E_1, E_2, \dots, E_n$ (with no intersection), $P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i)$.

Then, from the Axioms of Probability above, we can derive the following:

- **Definition 6.1.2.7:** The complement rule states that $P(\overline{E}) = P(S - E) = P(S) - P(E) = 1 - P(E)$.
- **Definition 6.1.2.8:** The impossible rule states that $P(\emptyset) = 0$.
- **Definition 6.1.2.9:** The general addition rule states that $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$.
- If $E_1 \subseteq E_2$, then $P(E_1) \leq P(E_2)$.

Example 6.1.2.1: A sequence of 10 bits is randomly generated. What is the probability that at least 1 of these is 0?

Solution: Let E be an event in which all bits are 1.

$$P(E) = \frac{|E|}{|S|} = \frac{1}{2^{10}}$$

Then, by the complement rule, $P(\overline{E}) = P(1 - E)$ where $\overline{E}$ is the event in which at least one bit is not 1.

$$P(\overline{E}) = 1 - \frac{1}{1024} = \frac{1023}{1024}$$

□

- 6.2 Dependency and Bayes Theorem
- 6.3 Discrete Probability Distributions
- 6.4 Continuous Probability Distributions
- 6.5 Gaussian Distribution and its Characteristics
- 6.6 Correlations
- 6.7 Central Limit Theorem and Sampling
- 6.8 Hypothesis Testing
- 6.9 Data Types and Visualisation
- 6.10 Basic Data Analysis with Python
- 7 Data Cleaning and Analytics

This section is taught by Dr Igor Razgon. [Check for recommended reading.](#)

8 Computer Graphics

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

9 Image Processing

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

Part III

Networks and Systems

10 Module Introduction

Networks and Systems has the module code COMP2211 and is led by Dr Wanqing Tu.

10.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 6. Additional formatives are also shown in *italics*.

Title	Opening	Due	Feedback	Professor	Weight
Cyber Security	9th Oct	5th Nov	3rd Dec	Dr Carlton Shepherd	25%
Networks	23rd Oct	21st Jan	18th Feb	Dr Wanqing Tu	25%
Dist. Sys. & Databases	May Exam	-	-	Dr Vasantam	50%

Table 6: Assessments for Networks and Systems.

10.2 Timetable

This module is taught through one two-hour lecture and one two-hour computer class each week as seen below in Table 7.

Day	Time	Class Type	Location	Lecturer
Thursday	09:00 - 11:00	Lecture	CLC202	Dr Carlton Shepherd
Thursday	16:00 - 18:00	Computer Class	CC-0007	-

Table 7: Michaelmas Term Timetable for Networks and Systems.

11 Cyber Security

This section is taught by Dr Carlton Shepherd. [Check for recommended reading.](#)

11.1 Introduction, Threat Modelling, and Applied Cryptography

12 Networks

This section is taught by Dr Wanqing Tu. [Check for recommended reading.](#)

13 Databases

This section is taught by Dr Vasantam. The recommended reading for this topic is:

- Connolly, T.M. and Begg, C.E. (2015). Database Systems: A practical approach to design, implementation, and management. 6th edn. Harlow, Essex, England: Pearson Education Limited.

- 13.1 Enhanced Entity-Relationship Model**
- 13.2 Semistructured Databases (XML)**
- 13.3 XML Data Manipulation (XPath and XQuery)**
- 13.4 Transactions**
- 13.5 Concurrency Control**
- 13.6 Distributed Transactions**
- 13.7 Distributed Concurrency Control**

14 Distributed Systems

This section is taught by Dr Vasantam. The recommended reading for this topic is:

- Van Steen, M. and Tanenbaum, A.S. (2025). Distributed Systems. 4th edn. [online] Self-Published. Available at: <https://www.distributed-systems.net/index.php/books/ds4/> [Accessed 5 Oct. 2026].

- 14.1 Introduction to Distributed Systems**
- 14.2 Middleware and RMI Technologies**
- 14.3 Message-Oriented Middleware and Web Services**
- 14.4 Replication Models**
- 14.5 Concurrent Events and Consistency Control**
- 14.6 Fault Tolerance**
- 14.7 Load Distribution**

Part IV

Programming Paradigms

15 Module Introduction

Programming Paradigms has the module code COMP2221 and is led by Dr Stuart James.

15.1 Assessment

This module is assessed through one piece of summative coursework and two final exams, these can be seen in Table 8. Additional formatives are also shown in *italics*.

Title	Opening	Due	Feedback	Professor	Weight
<i>Systems Programming</i>	<i>12th Oct</i>	<i>12th Nov</i>	<i>10th Dec</i>	<i>Dr Stuart James</i>	-
Systems Programming	23rd Oct	10th Dec	21st Jan	Dr Stuart James	50%
Object Oriented & Functional	May Exam	-	-	Ivkovic & Moses	50%

Table 8: Assessments for Programming Paradigms.

15.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 9.

Day	Time	Class Type	Location	Lecturer
Monday	11:00 - 12:00	Lecture	CLC202	Dr Stuart James
Monday	16:00 - 18:00	Computer Class	CB-LG001	n/a
Friday	14:00 - 15:00	Lecture	CLC202	Dr Stuart James

Table 9: Michaelmas Term Timetable for Programming Paradigms.

16 Systems Programming

This section is taught by Dr Stuart James. There is no official reading list for this module.

Definition 16.0.0.1: Systems programming involves the development of individual pieces of software that allow the entire system to function as a single unit.

16.1 Version Control and UNIX

16.2 The Basics of C

16.2.1 Basic File Setup

Definition 16.2.1.1: Pre-processor directives are lines which start with a `#` and are commands to the C pre-processor.

- If the directive is `#include`, and `<>` are used, then `/usr/include` is prioritised in UNIX by convention, whereas if `"` are used, then the current directory is prioritised.
- **Definition 16.2.1.1:** If the directive is `#define`, then this is a definition macro. These are used to provide definitions in code.
- **Definition 16.2.1.2:** You can also use directives for debugging without generating overheads, these are called conditionals. They are as shown in Table 10.

Conditional	Function
<code>#ifdef VAR</code>	Tests if VAR is <code>#defined</code> .
<code>#ifndef VAR</code>	Tests if VAR is not <code>#defined</code> .
<code>#else</code>	Runs if the original condition evaluates as false.
<code>#endif</code>	Ends the if testing condition.

Table 10: Different types of conditionals.

You can also put debugging code in the preamble, as seen in Algorithm 1.

```
#define MY_DEBUG // Define an identifier.
```

```
#ifdef MY_DEBUG
    assert(i > 0);
    printf("i is %d \n", i);
#endif
```

Algorithm 1: Debugging Code Example.

Definition 16.2.1.4: You can also use `#define` to define parameterised macros. These act like functions, and an example can be seen in Algorithm 2.

```
#define ADD(a,b) a + b
```

Algorithm 2: Parameterised Macro Example.

Definition 16.2.1.5: The main() function is the entry function for C programs. You can only have one of these.

16.2.2 Printf Parameters

Value	Resulting Format
<code>%d</code>	Signed decimal (int).
<code>%u</code>	Unsigned decimal.
<code>%o, %x</code>	Octal and hexadecimal.
<code>%l</code>	Long.
<code>%f</code>	Floating point, so <code>%1.2f</code> will give 3.14.
<code>%e</code>	Floating point (exponent form).
<code>%c, %s</code>	Character and string.

Table 11: First parameter of printf.

16.3 Control Flow and Functions

16.4 Data Types and Structures

16.5 Advanced Bash Scripting

16.6 Pointers

16.7 Memory Management

16.8 Debugging

16.9 Scope and Recursion

16.10 Libraries

16.11 stdlib

16.12 C++

17 Functional Programming

This section is taught by Dr Billy Moses Jr.. [Check for recommended reading.](#)

18 Object Oriented Programming

This section is taught by Dr Mladen Ivkovic. [Check for recommended reading.](#)

18.1 Java Fundamentals

18.2 OOP with Classes and Objects

18.3 Encapsulation

18.4 Inheritance

18.5 Abstraction with Abstract Classes

18.6 Abstraction with Interfaces

18.7 Polymorphism

18.8 Java Collections

18.9 Best Practises

Part V

Software Engineering

19 Module Introduction

Software Engineering has the module code COMP2281 and is led by Dr Craig Stewart.

19.1 Assessment

This module is assessed through a group project, the deadlines for this can be seen in Table 12. Additional formatives are also shown in *italics*.

Title	Opening	Due	Feedback	Professor	Weight
Peer Evaluation 1	16th Oct	19th Nov	10th Dec	Dr Craig Stewart	1.66%
Requirement Documentation	16th Oct	19th Nov	17th Dec	Dr Craig Stewart	15%
Peer Evaluation 2	16th Oct	11th Feb	11th Mar	Dr Craig Stewart	1.67%
Product Presentation	Week 16	-	18th Mar	Dr Craig Stewart	15%
Technical Report	5th Feb	18th Mar	6th May	Dr Craig Stewart	30%
Peer Evaluation 3	5th Feb	18th Mar	6th May	Dr Craig Stewart	1.67%
Git Commit	5th Feb	29th Apr	-	Dr Craig Stewart	5%
Project Handover	5th Feb	29th Apr	-	Dr Craig Stewart	10%
Reflective Experience Report	5th Feb	6th May	-	Dr Craig Stewart	20%

Table 12: Assessments for Software Engineering.

19.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 13.

Day	Time	Class Type	Location	Lecturer
Tuesday	09:00 - 10:00	Lecture	TLC033	Dr Nelly Bencomo
Wednesday	11:00 - 12:00	Lecture	TLC033	Dr Nelly Bencomo
Thursday	11:00 - 13:00	Practical	CC-0007	Dr Gagangeet Aujla

Table 13: Michaelmas Term Timetable for Software Engineering.

Add lectures into relevant sections when you do them with the lecturer linked. Check for recommended reading.

19.3 Introduction to the Module

Listen to the recording of lecture 1.

20 Project Management

21 Software Engineering

22 Legal, Social, Ethical, and Professional

23 Group Project

Part VI

Theory of Computation

24 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

24.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 14. Additional formatives are also shown in italics.

Title	Opening	Due	Feedback	Professor	Weight
Benchtest 1	Week 11	-	2nd Feb	Dr Dantchev and Dr Mertzios	12.5%
Benchtest 2	Week 20	-	-	Dr Dantchev and Dr Mertzios	12.5%
Final Exam	May Exam	-	-	Dr Barnaby Martin	75%

Table 14: Assessments for Theory of Computation.

24.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 15.

Day	Time	Class Type	Location	Lecturer
Tuesday	10:00 - 11:00	Lecture	CLC202	Dr Barnaby Martin
Wednesday	09:00 - 11:00	Problem Class	TLC116	-
Thursday	14:00 - 15:00	Lecture	CLC202	Dr Sukanya Pandey

Table 15: Michaelmas Term Timetable for Theory of Computation.

25 Models of Computation

This section is taught by Dr Barnaby Martin. [Check for recommended reading.](#)

25.1 Finite-State Automata and Regular Languages

25.1.1 Formal Definitions

Definition 25.1.1.1: A deterministic finite-state automaton is a five-tuple $(Q, \Sigma, \delta, q_0, F)$ where Q is the finite set of states, Σ is a finite alphabet, $\delta : Q \times \Sigma \rightarrow Q$ is the transition function, $q_0 \in Q$ is the start state, and $F \subseteq Q$ is the set of accept states.

Definition 25.1.1.2: A DFA M recognises the language L if $L = \{w | M \text{ accepts } w\}$.

Definition 25.1.1.3: A language is called a regular language if some DFA recognises it.

Definition 25.1.1.4: A regular expression R defines a language $L(R)$. We shall eventually prove that $RE \equiv DFA$. R is a regular expression over the alphabet Σ if R is:

1. a for some $a \in \Sigma$.
2. ϵ (the empty string).
3. $\emptyset$ (the empty set).

4. $(R_1 \cup R_2)$ where R_1 and R_2 are regular expressions.
5. $R_1 \cdot R_2$ where R_1 and R_2 are regular expressions.
6. R_1^* where R_1 is a regular expression.

The list of regular operations can be seen in Table 16 where the operations labelled with $^+$ are language theory specific.

Operation Name	Notation	Set Equivalent
Union	$A \cup B$	$\{x x \in A \text{ or } x \in B\}$
Intersection	$A \cap B$	$\{x x \in A \text{ and } x \in B\}$
Difference	$A \setminus B$	$\{x x \in A \text{ and } x \notin B\}$
Complement	$\overline{A}$	$\Sigma^* \setminus A$
Concatenation ⁺	$A \circ B$	$\{xy x \in A \text{ and } y \in B\}$
Star ⁺	A^*	$\{x_1 x_2 \dots x_k k \geq 0 \text{ and } x_i \in A \text{ for every } i, 1 \leq i \leq k\} \equiv \bigcup_{i=0}^{\infty} A^i$

Table 16: Regular Operations.

25.1.2 Combining Automata

Sometimes we may want to implement an operation on multiple languages. We can do this by combining their automata. If we have that L_1 is recognised by $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ and L_2 is recognised by $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$; we want to combine M_1 and M_2 into a new automaton M that would recognise $L_1 \cup L_2$.

You would think that we could just run the automata sequentially, and accept if either M_1 or M_2 accept, but this does not work. This is because after M_1 has run on the input, the input is exhausted and there is no way to rewind it in order to run M_2 .

Instead, we can attempt to run M_1 and M_2 on the input in parallel. We do this by using pairs of states (one for each DFA). Formally, the components of the new automaton $M = (Q, \Sigma, \delta, q_0, F)$ are:

1. The set of possible states contains every possible cross-set pair of Q_1 and Q_2 .

$$Q = Q_1 \times Q_2$$

2. For every $s \in Q_1, u \in Q_2, a \in \Sigma$. The transition function contains every transition from a pair of states from Q being given any input from Σ .

$$\delta((s, u), a) = (\delta_1(s, a), \delta_2(u, a))$$

3. The start state is the pair containing both start states from M_1 and M_2 .

$$q_0 = (q_1, q_2)$$

4. The set of accept states is every possible cross-set pair of accept states from F_1 and F_2 .

$$F = \{(s, u) \in Q | s \in F_1 \text{ or } u \in F_2\}$$

25.1.3 Non-Deterministic Finite Automaton

If, instead you wish to work with concatenation, such that $w \in L_1 \circ L_2$ only if there are words x and y such that $w = xy$, $x \in L_1$ and $y \in L_2$. You need to run M_1 on a prefix of w and then run M_2 on the rest. This must be done non-deterministically and stops becoming a DFA as all accept states of M_1 have to map to the start state of M_2 .

Definition 25.1.3.1: A non-deterministic finite automaton is a five-tuple $(Q, \Sigma, \delta, q_0, F)$, where Q is the finite set of states, Σ is a finite alphabet, $\delta : Q \times (\Sigma \cup \{\epsilon\}) \rightarrow \mathcal{P}(Q)$ is the transition function, $q_0 \in Q$ is the start state, and $F \subseteq Q$ is the set of accept states.

Computing NFA

Before computing an NFA, an important convention to note is that a single step in a computation consists of first reading the next input symbol followed by taking any number of ϵ -transitions. This means that empty strings ϵ s can be freely added inside the actual word $\in \Sigma^*$, e.g., 010110 can be viewed as, say, $\epsilon 01\epsilon\epsilon 01\epsilon 01$.

Let $M = (Q, \Sigma, \delta, q_0, F)$ be an NFA and $w = w_1w_2\dots w_n$ be a word over $\Sigma \cup \{\epsilon\}$. M accepts w if there exists a sequence of states $r_0, r_1, r_2, \dots, r_n$ satisfying the following conditions:

1. $r_0 = q_0$.
2. $r_{i+1} \in \delta(r_i, w_{i+1})$ for every $i, 0 \leq i \leq n - 1$.
3. $r_n \in F$.

25.1.4 Converting NFA into DFA

The difference between a DFA and NFA is that a DFA has a unique computation (path) and either accepts or rejects, while a NFA generally has many computation paths and accepts if at least one computation accepts and only rejects if all the computations reject.

Reached slide 22/55 (powerpoint says slide 21).

25.2 Büchi Automata and Omega-Regular Languages

25.3 Linear-Time Logic and Translating Into Büchi Automata

26 Algorithms and Complexity

This section is taught by Dr Sukanya Pandey. The recommended reading for this topic is:

- Introduction to Algorithms (Cormen T.H., et al., 2009).

26.1 Asymptotic Complexity and Notation

26.2 Greedy Algorithms

26.3 Divide and Conquer Algorithms

26.4 Dynamic Programming Algorithms

26.5 Matchings in Graphs

Part VII

Appendix

27 Artificial Intelligence

27.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: Monday 15:30 - 16:30 in Term 1.

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00 in Term 2.

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: Thursday 14:30 - 15:30 in Term 2.

27.2 References

27.3 List of Figures

27.4 Reading List

27.4.1 Machine Learning

Long, Y. and Durham University (2026). COMP2261.Machine.Learning.Study.and.Coursework.Guide-with.Navigation.pdf.

Deisenroth, M.P., Faisal, A.A. and Ong, C.S. (2020). Mathematics for Machine Learning. [online] Cambridge University Press. Available at: <https://mml-book.com/> [Accessed 3 Oct. 2026].

Hackeling, G. (2017). Mastering Machine Learning with scikit-learn. 2nd edn. Packt Publishing.

27.4.2 Ethics and Bias in AI

Various, will be added over time.

28 Data Science

28.1 Module Professors

Module Lead: Professor Thomas Erlebach

- Focus: Algorithms and networks.
- Email: thomas.erlebach@durham.ac.uk
- Office Hours: n/a.

Lecturer: Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00.

Lecturer: Doctor George Alex Koulieris

- Focus: Image processing and computer graphics.
- Email: georgios.a.koulieris@durham.ac.uk
- Office Hours: Please email to make an appointment.

28.2 References

28.3 List of Figures

28.4 Reading List

28.4.1 Machine Learning

Degroot, M.H. and Schervish, M.J. (2012). Probability and Statistics. Boston Mass.: Pearson.

29 Networks and Systems

29.1 Module Professors

Module Lead: Doctor Wanqing Tu

- Focus: Computer networks and wireless communications.
- Email: wanqing.tu@durham.ac.uk
- Office Hours: Thursday 11:00 - 12:00.

Lecturer: Doctor Thirupathaiah Vasantam

- Focus: Applied probability and performance modelling.
- Email: thirupathaiah.vasantam@durham.ac.uk
- Office Hours: Monday 14:00 - 15:00.

Lecturer: Doctor Carlton Shepherd

- Focus: Cyber security.
- Email: carlton.g.shepherd@durham.ac.uk
- Office Hours: Email to make an appointment.

29.2 References

29.3 List of Figures

29.4 Reading List

29.4.1 Databases

Connolly, T.M. and Begg, C.E. (2015). Database Systems: A practical approach to design, implementation, and management. 6th edn. Harlow, Essex, England: Pearson Education Limited.

29.4.2 Distributed Systems

Van Steen, M. and Tanenbaum, A.S. (2025). Distributed Systems. 4th edn. [online] Self-Published. Available at: <https://www.distributed-systems.net/index.php/books/ds4/> [Accessed 5 Oct. 2026].

30 Programming Paradigms

30.1 Module Professors

Module Lead: Doctor Stuart James

- Focus: Artificial intelligence, computer vision, human computer interaction, and digital humanities.
- Email: stuart.a.james@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Billy Moses Jr.

- Focus: Algorithms and biologically inspired computing.
- Email: william.k.moses-jr@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Mladen Ivkovic

- Focus: Performance prediction and optimisation.
- Email: mladen.ivkovic@durham.ac.uk
- Office Hours: Email to make an appointment.

30.2 References

30.3 List of Figures

31 Software Engineering

31.1 Module Professors

Module Lead: Doctor Craig Stewart

- Focus: Adaptive systems, HCI, and user modelling.
- Email: craig.d.stewart@durham.ac.uk
- Office Hours: Friday 11:00 - 12:00.

Lecturer: Doctor Nelly Bencomo

- Focus: Software engineering for autonomous, self-adaptive, AI and ML systems.
- Email: nelly.bencomo@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Gagangeet Aujla

- Focus: Data-driven and data-intensive cyber-physical systems and digital twins.
- Email: gagangeet.s.aujla@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Gokberk Kabacaoglu

- Focus: Machine learning and numerical applications.
- Email: gokberk.kabacaoglu@durham.ac.uk
- Office Hours: Thursday 14:00 - 15:00.

Lecturer: Doctor Ryan Crosby

- Focus: The assessment of novice programmers.
- Email: ryan.crosby@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Sukanya Pandey

- Focus: Graph theory.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Email to make an appointment.

31.2 References

31.3 List of Figures

32 Theory of Computation

32.1 Module Professors

Module Lead: Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.
- Email: barnaby.d.martin@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Sukanya Pandey

- Focus: Graph theory.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: Flexible, contact to arrange.

Lecturer: Doctor Stefan Dantchev

- Focus: Combinatorics.
- Email: s.s.dantchev@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor George Mertzios

- Focus: Algorithms and networks.
- Email: george.mertzios@durham.ac.uk
- Office Hours: **To be confirmed.**

32.2 References

32.3 List of Figures

32.4 Recommended Reading

32.4.1 Algorithms and Complexity

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C. (2009). Introduction to Algorithms. 3rd edn. MIT Press.